

The Effect of Boarding Strategies on Aircraft Boarding Efficiency

A Simulation-Based Comparative Study | Author: Zihao Huang | October 2026

Abstract

This study builds a simplified single-aisle cabin boarding model to compare the total boarding time of three strategies: *random*, *back-to-front*, and *zoning (5 zones)*. Contrary to intuition, the simulation shows that **random boarding is the fastest and back-to-front is the slowest**, and the gap widens as stow time and the number of rows increase. A sensitivity analysis over stow time and cabin size confirms the result is stable. The mechanism is that random boarding lets passengers in different rows stow luggage **simultaneously (in parallel)**, whereas back-to-front forces passengers of the same row to stow **sequentially**, under-using the aisle. This indicates that "organized boarding" is not necessarily faster and provides a quantifiable framework for airline operations optimization.

1. Background and Research Question

Boarding time strongly affects on-time performance and operating cost. Airlines often adopt "zone boarding" or "back-to-front boarding", but whether these outperform random boarding is debated. This study tests the following hypothesis:

Hypothesis H: Organized strategies (back-to-front / zoning) board faster than random boarding.

2. Methods

2.1 Model

- The cabin has a **single aisle**; passengers enter from the front and walk to their assigned row.
- On reaching their row, a passenger **stows luggage**, occupying the aisle for **STOW** time steps (ticks); followers cannot pass.
- Each passenger advances at most one cell per tick; if blocked, they wait (no overtaking).
- Only **one passenger enters** per tick (single-file boarding).
- The simulation ends when all passengers are seated; the total number of ticks is the **boarding time** (smaller = faster).

2.2 Variables

Type	Variable
Independent	Boarding strategy (Random / Back-to-front / Zones-5)
Dependent	Total boarding time (ticks)
Control (baseline)	Random boarding (normalized to 1.00)
Controlled	Rows R; seats per row (6); entry rate (1/tick); repeats (20)

2.3 Data Collection

Strategies with randomness (Random, Zones) were each run **20 times and averaged** to reduce random error; Back-to-front is deterministic and requires a single run. Results are reported as "strategy time ÷ random time" (baseline = 1.00).

3. Results

3.1 Main Result (R = 30 rows, STOW = 3)

Strategy	Average boarding time (ticks)	Relative to random
Random	331.9	1.00 (baseline)
Zones (5)	402.6	1.21
Back-to-front	627	1.89

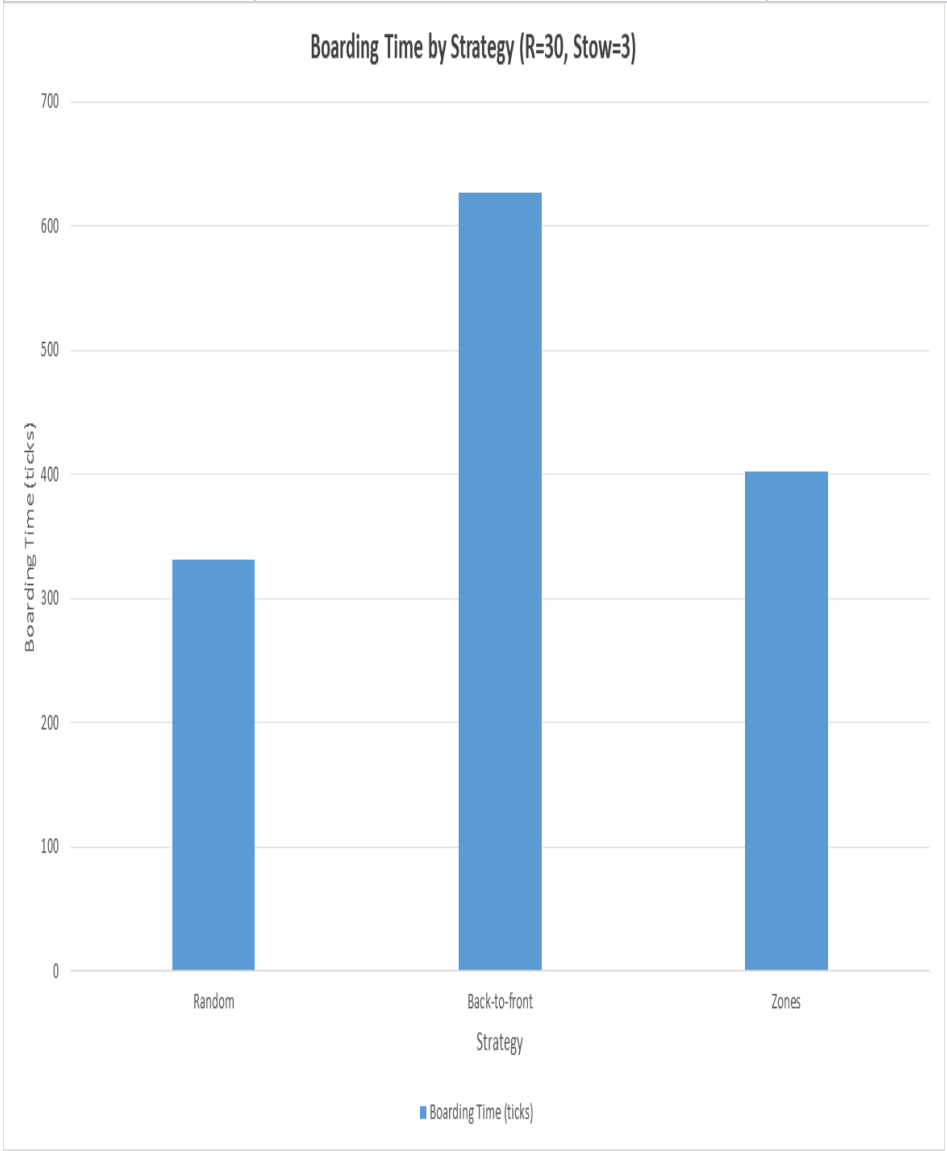


Figure 1. Boarding time by strategy (R=30, STOW=3): random is fastest, back-to-front slowest.

3.2 Sensitivity Analysis

(a) Varying stow time (fixed R = 30). The disadvantage of back-to-front grows sharply with stow time.

STOW	Random	Back-to-front	Zones (5)	Rel. random (back)	Rel. random (zones)
1	238.9	325	249.9	1.36	1.05
3	331.9	627	402.6	1.89	1.21
6	471.5	1080	633.8	2.29	1.34

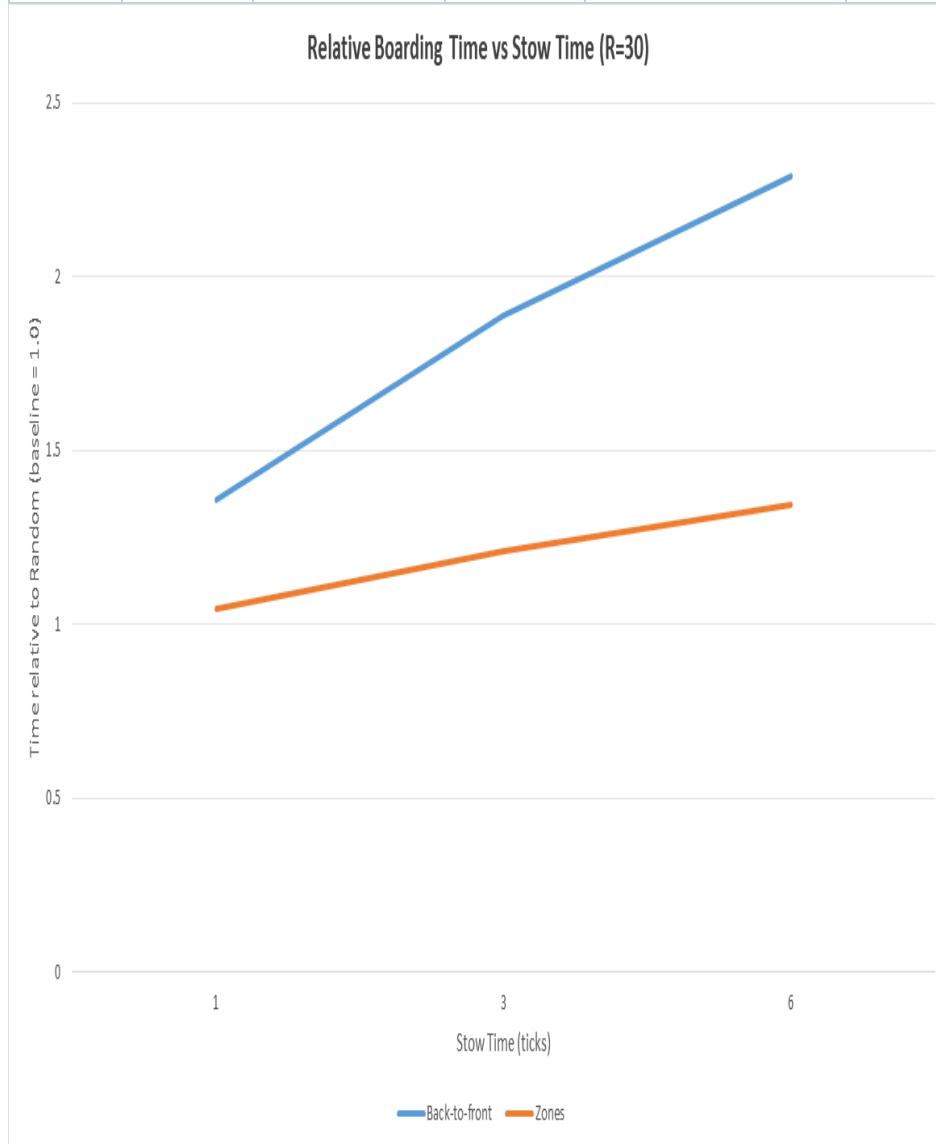


Figure 2. Relative boarding time vs stow time (R=30): back-to-front worsens steadily; zoning stays nearly flat.

(b) Varying number of rows (fixed STOW = 3). Larger aircraft penalize back-to-front more.

Rows R	Random	Back-to-front	Zones (5)	Rel. random (back)	Rel. random (zones)
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20	234.3	417	288.7	1.78	1.23
30	331.9	627	402.6	1.89	1.21
40	423.6	837	507.3	1.98	1.20

4. Discussion

Why is random fastest? Random boarding spreads passengers across different rows, so several rows stow luggage **simultaneously**, using the aisle in parallel and maximizing throughput.

Why is back-to-front slowest? It boards the six passengers of the same row consecutively; they all must stow at the same aisle position, so they can only proceed **one by one (sequentially)**. Meanwhile, rear passengers walk the full aisle, amplifying congestion. Hence the disadvantage grows with stow time (STOW) and cabin size (R).

Deviation from intuition shows that "organized" does not mean "efficient". This is consistent with parts of the real boarding literature, where back-to-front is often shown to be sub-optimal while random or finer seat-ordered methods (e.g., the Steffen method) tend to be more efficient.

5. Limitations and Future Work

- The model is simplified: it ignores assigned seating effects, dual aisles, two doors, and passengers stepping into seats immediately after stowing.
- Future work could add "window-middle-aisle" and the "Steffen method" for comparison.
- The entry rate and parallel-boarding assumptions could be varied to test robustness.

6. Conclusion

In this simulation, **dispersed boarding (random) outperforms clustered boarding (back-to-front)**, because parallel luggage stowing saves more time than sequential waiting; the disadvantage of back-to-front increases with stow time and cabin size. These findings are relevant to the design of boarding strategies in airline operations.

References

Steffen, J. H. (2008). Optimal boarding method for airline passengers. *Journal of Air Transport Management*. Van Landeghem, H., & Beuselinck, A. (2002). Reducing passenger boarding time in airplanes. *European Journal of Operational Research*. Bachmat, E., et al. (2006). Analysis of airplane boarding via space-time geometry and random matrix theory.